

Matplotlib is a visualization library in Python used in machine learning to present data in charts, lines, bars, and other graphical formats.

NumPy is a Python-based library used for scientific and numerical calculations. It allows performing mathematical operations on n-dimensional arrays.

Data Science is the process of extracting knowledge and insights from structured and unstructured data.

Data Cleaning is the process of detecting and removing errors and inconsistencies from input data. It improves data quality and is commonly done using Pandas.

Pandas is a Python library used in data science for loading data from sources like Excel and CSV files and for cleaning and manipulating data.

Data from an Excel file can be loaded using pandas by writing: `import pandas as pd` and `df = pd.read_excel("file_path")`.

Missing data can be handled using `fillna()` and `dropna()` functions. `fillna()` replaces missing values, while `dropna()` removes rows or columns with missing values.

Imputing is the process of filling missing values using strategies like mean.

A machine learning project involves loading data, cleaning it, splitting it into training and testing sets, training the model, and testing the model.

The model is trained using the `fit()` method with training data.

The model is tested using the `predict()` method with testing data.

Model accuracy for regression is measured using the `score()` method, while for classification it is measured using `accuracy_score()`.

A confusion matrix is used to evaluate classification performance, where higher true positive and true negative values indicate better accuracy.

Regression model efficiency can be measured using Mean Squared Error (MSE).

Machine Learning is the ability of a system to learn automatically without explicit programming.

Machine learning works by training algorithms with data and then using them to make predictions.

Supervised learning uses labeled data and is used for classification and regression problems, while unsupervised learning uses unlabeled data and is used for clustering and pattern detection.

Common regression algorithms include Linear Regression, Lasso, Ridge, and Support Vector Regression.

Common classification algorithms include Naive Bayes, Support Vector Machine, Decision Tree, Random Forest, and Logistic Regression.

Deep Learning is a subset of machine learning that uses neural networks for predictions.

Machine learning works well with smaller datasets and requires less computational power, while deep learning works best with large datasets and requires high computational power like GPUs.

Neural Networks are systems inspired by the human brain, consisting of layers of interconnected neurons.

A neural network has input, hidden, and output layers, and data flows through these layers to generate predictions.

A neuron is the smallest unit of a neural network. It receives input, applies weights, sums them, and passes the result through an activation function.

CNN (Convolutional Neural Network) is a deep learning algorithm mainly used for image detection.

CNN consists of convolution, ReLU, pooling, and fully connected layers.

CNN works by detecting features in different parts of an image.

NLP (Natural Language Processing) is a branch of AI that enables computers to understand and process human language.

NLTK (Natural Language Toolkit) is a Python library used for natural language processing tasks.